DCLab R&D; Lab and Research Studio

A simple, evidence grounded explanation of the system, its agents, its experiments, and its results

The 30 second explanation

DCLab's main product aims to help people build and operate machine learning work in one connected project. This R&D repository is its test workshop. It runs model building experiments on real datasets, records what worked and failed, and turns the evidence into reusable workflow guidance. The Research Studio adds AI agents that propose and explain bounded experiments while tested code calculates the scores.

Prepared from the repository's reports, result JSON, normalized registry, and local Research Studio ledger. Results are as recorded on 25 September 2026. This is a reading guide to evidence, not a claim that a benchmark model is ready for real users.

Evidence: S1, S2, S3, S4, S5

How to read this guide

If you want to know...	Read
What is DCLab and why does the lab exist?	Sections 1-2
How do agents and experiments work?	Sections 3-5
What do the numbers mean?	Sections 6-8
What exactly happened in each campaign?	Appendices A-F
What can be reused or fine tuned?	Sections 9-10
Where are the underlying records?	Sources and evidence map

Read the label next to a number

Historical registry scores, the controlled 50 experiment holdouts, churn development CV, and agentic development CV use different protocols. A higher number from one group does not prove its model is better than a number from another group.

1. What is DCLab?

DCLab is being built as a project centered environment for data scientists and ML engineers. Its intended path connects the question, data, experiments, model decisions, release, and monitoring. A developer should be able to see what was tried, why, what the tests found, and what can safely happen next. The main product and this R&D repository are separate codebases.

Part	Simple meaning	Current relationship
DCLab main product	The workspace where users would build and operate real ML projects.	Its current truth and implementation live in the separate DCLab product repository.
R&D Lab	The workshop for repeatable research on datasets, algorithms, features, leakage, and evaluation.	This repository contains the experiments and evidence reviewed here.
Research Studio	A local interface and agent loop for asking questions, running a small experiment series, and inspecting results.	Implemented inside this R&D repository at 127.0.0.1:8765.
Knowledge memory	Rules and examples learned from research, each linked to evidence.	Useful input to future product decisions and possible SLM training after review.

2. What is this lab trying to learn?

The lab asks practical questions: What do the columns mean? Which fields would exist when a prediction is made? Which features help and which create leakage? Does a simple linear model beat a tree? Does tuning produce a repeatable improvement? What happens if a field is missing? Which steps should be repeated on a new dataset?

- A good lab result is a narrow lesson with a source file, dataset, split, metric, limitation, and next test.
- A failed or worse experiment is still valuable: it tells the next researcher which idea needs revision.
- The longer term goal is a reusable workflow, not a single universal best model.

Evidence: S1, S2, S4, S12

3. What exists today?

Evidence collection	Size	What it answers	Important limit
Historical normalized registry	579 records / 11 datasets / 7 suites	Which methods were tried across the older notebooks and runners?	Protocols differ; only 10 records have full provenance. Two duplicate key warnings remain.
Controlled model building campaign	50 tasks = 10 public datasets x 5 stages	How does the complete data-to-model workflow behave under one campaign protocol?	Ten final holdouts were used once. Cached category labels need source repair.
HyperAck core lifecycle	83 records = 15 original + 15 safe + 53 optimized safe; 80 completed, 3 skipped	What changes when post-outcome fare fields are excluded?	Historical split was consulted repeatedly; fresh temporal confirmation is needed.
HyperAck four way benchmark	44 additional registry rows	How do safe/unsafe and baseline/optimized combinations compare?	It overlaps the same problem and does not add 44 independent datasets.
Telco Churn fixed suite	15 completed recipes	Which model and charge/tenure features look useful?	Development CV only; no independent confirmation.
Agentic Research Studio	5 run records	Can agents propose, execute, critique, and remember bounded tests?	Adaptive development CV; failed and budget-limited runs are retained.

The counts overlap. For example, the 579-row registry includes the 127 HyperAck rows and ten final records from the 50-task campaign. Do not sum all rows and call them independent experiments.

Evidence: S2, S3, S4, S5, S6, S7

4. How does the Research Studio work?

A person enters a goal, selects a project and dataset, and sets limits on experiments, rows, CV repeats, active minutes, and LLM calls. The backend stores a durable run ID. A graph coordinates specialist agents and a separate worker calculates the actual measurements.

Order	Actor	What happens	Output
1	Deterministic profiler	Reads an allowed data snapshot and summarizes rows, columns, missingness, duplicates, and target balance.	Aggregate profile and data fingerprints.
2	Research Planner	Turns the user's goal into bounded questions, sequence, success criteria, and limitations.	Typed agenda.
3	Data Scientist	Reviews the profile, feature availability assumptions, leakage risks, and sensible feature hypotheses.	Typed data review.
4	Experiment Designer	Proposes one allowed model and feature configuration at a time, citing prior evidence when comparing.	Validated experiment recipe.
5	Deterministic worker	Applies decision-time feature policy, runs fold-local preprocessing, fits models, and computes scores and diagnostics.	Result JSON, OOF predictions, replay recipe, hashes.
6	Results Critic	Checks whether the measured change answers the question and states limitations and next test.	Critique with evidence IDs.
7	Knowledge Curator	After the bounded loop, turns supported results and failures into narrow lessons and workflow blocks.	Provisional synthesis.

Who owns the truth?

Agents propose and interpret. The validated worker computes metrics. SQLite and on-disk files preserve the run. LangGraph coordinates and checkpoints the sequence; NOOA supplies typed agent classes. A human must still review business meaning, sensitive use, and any production decision.

Evidence: S8, S9, S10

5. How is one experiment made?

The designer must submit a structured recipe: dataset, testable hypothesis, one of seven allowed model families, bounded model parameters, optional safe feature formulas, columns to drop, fields to simulate as missing, and cited earlier results. It cannot submit arbitrary Python to execute.

Step	Plain explanation	Technical check
Define prediction moment	When would the answer be needed?	Block fields that appear only after the outcome, such as HyperAck final fares.
Choose a fair split	Keep lookalike copies from appearing in both study and quiz.	Exact duplicate allowed-input rows share a group in repeated 3-fold CV.
Prepare inside each fold	Learn filling and category rules from study rows only.	Imputer, scaler, encoder, derived features, and estimator are in one pipeline.
Train and predict	Fit on two folds, predict on the third, repeat.	Default is 2 repeats x 3 folds; collect out-of-fold probabilities.
Compare	Ask whether the new choice helped the same data splits.	Paired fold delta is descriptive, not a significance test.
Challenge	Ask what would break at use time.	Calibration, missing-input stress, sensitivity, and feature lineage diagnostics.
Save and critique	Keep the result and the explanation together.	Evidence ID, result, recipe, source/code hashes, critique, failure if any.

A concrete example

For Telco Churn, one agentic run first tested logistic regression on existing account features. The next recipe added a log transform of tenure while holding the model family fixed. Mean development CV ROC-AUC moved from 0.8447 to 0.8483; the paired mean gain was about 0.0036. A later histogram boosting test scored 0.8214. The sequence illustrates an interpretable ablation and a useful negative result. It does not show that logistic regression will win on future customers.

Evidence: S5, S8, S9

6. What do the numbers mean?

Metric	Simple meaning	Better direction	Common trap
ROC-AUC	How often a random positive case ranks above a random negative case.	Higher	It does not choose a threshold or prove useful business decisions.
Average precision (AP)	Precision across recall levels; useful when positives are rare.	Higher	Different class balance can change AP, so cross-dataset comparison is unfair.
Log loss	Penalty for confident wrong probabilities.	Lower	A model may improve AUC while worsening probability quality.
Brier score	Average squared error of predicted probabilities.	Lower	It mixes discrimination and calibration.
Recall	Share of actual positives found at a chosen threshold.	Higher, subject to cost	Can rise by flagging almost everyone; inspect precision too.
F1	Balance of precision and recall at one threshold.	Higher, subject to cost	Ignores true negatives and business costs.
Paired delta	New score minus reference score on matching folds.	Positive for AUC/AP	Small deltas from reused CV can be noise or adaptive overfitting.

How to read 0.85 AUC

It suggests useful ranking in that evaluation setup. It does not mean 85% accuracy, 85% precision, or an 85% chance that each individual prediction is correct.

7. What have the results taught us?

Finding	Measured support	What to do with it
Leakage can dominate a leaderboard	Historical safe/unsafe AUC gaps: bank +0.1328, online shoppers +0.1575, HyperAck +0.0347. The controlled campaign found bank +0.1842 and online shoppers +0.1968 on its own CV protocol.	Define the prediction moment and audit timestamps/lineage before model comparison.
More features are not always better	The controlled campaign retained raw features on 7/10 datasets; ratios on two and interactions on one.	Start with raw inputs. Keep a feature recipe only when paired evidence and operational meaning justify it.
No model family wins everywhere	LightGBM led mean historical rank; Extra Trees was selected on 7/10 controlled datasets; churn's best fixed suite used engineered logistic regression.	Screen a linear baseline and a few nonlinear families on identical folds.
Input availability matters	The worker measures the effect of masking selected input columns; history/price fields still need real availability contracts.	Treat feature importance as a review candidate, not approval.
High benchmark scores need scrutiny	Mushroom reached 1.0000 controlled holdout AUC; breast cancer reached 0.9960.	Audit task ease, duplicates, proxies, split design, and intended use before claiming success.

These are scoped findings, not universal laws. The 22 machine readable rules in this repository attach confidence, evidence, exceptions, and next tests to each principle.

Evidence: S2, S3, S4, S11

8. The three projects in the Studio

Project	Question	Results to remember	Limit
Open tabular lab	How well does the five stage model building procedure transfer across ten public classification datasets?	50/50 controlled tasks complete. Raw features selected 7/10 times; Extra Trees 7/10 model screens.	Legacy cached categories and public benchmark splits limit real-world claims.
HyperAck	Can an order-time delivery acceptance model be accurate without final fare fields?	Unsafe historical AUC roughly 0.9802; best optimized safe AUC roughly 0.9455. The 0.0347 gap illustrates leakage severity.	Historical test use is adaptive; later-time, zone, and demand regime tests remain.
Telco Churn	Can account and billing data help rank likely churn?	Fixed 15 run leader CHURN-014: AUC 0.8499, AP 0.6708; agentic log-tenure test: development AUC 0.8483 on its separate 4,000-row protocol.	No independent churn holdout or verified outcome window.

HyperAck: what safe and unsafe mean

The unsafe tests used fields such as final_customer_fare and final_biker_fare, which are known only after the decision point in the order-time prediction story. Their high scores are useful for teaching leakage, not for selecting a deployable model. Safe tests exclude these fields. The small gain from extensive safe optimization also shows why a complicated model must justify its added cost.

Churn: why a simple model can win

A tenure/charge representation made logistic regression slightly stronger than the other tested families in the fixed suite. This is a dataset-specific development result; the agentic run used different row limits and split details, so its 0.8483 AUC is not a replication of the fixed suite's 0.8499 AUC.

Evidence: S2, S3, S4, S5, S7

9. The reusable ten step model building workflow

These blocks are a guide for an engineer or a focused future agent. They are ordered so that each decision has the information it needs.

ID	Step	What to check before moving on
WF-01	Prediction contract	define entity and decision -> define prediction moment -> define target window -> define action and costs -> freeze allowable information
WF-02	Source and lineage gate	inventory sources -> hash immutable snapshot -> record event timestamps -> map raw-to-modeled fields -> label owner and SLA
WF-03	Split design	identify independence unit -> choose group/time/domain split -> lock test IDs -> verify no overlap -> version split manifest
WF-04	Train-only EDA	schema and type audit -> target and missingness profile -> duplicates/entities -> distribution slices -> association hypotheses without promotion
WF-05	Leakage audit	semantic availability matrix -> lineage/timestamp check -> target-proxy review -> split/preprocessing audit -> safe-vs-unsafe severity ablation
WF-06	Feature ladder	raw baseline -> single domain transform -> ratios/differences -> bounded interactions -> external point-in-time features -> retain only reproducible lift
WF-07	Algorithm screen	dummy floor -> regularized linear model -> bagged trees -> boosting families -> domain-appropriate specialist -> rank on identical folds
WF-08	Conservative optimization	freeze objective -> small explicit search -> paired fold comparison -> proper scoring and runtime -> accept or keep baseline
WF-09	Reliability challenge	calibration -> threshold/cost analysis -> missing-input stress -> subgroup/slice floors -> temporal/domain holdout -> failure and rollback rule
WF-10	Knowledge capture	observed result -> narrow interpretation -> limitations/counterexample -> evidence paths and hashes -> next discriminating test -> human review status

Why this is a workflow rather than a checklist

A result from a later step can send the team back: a suspiciously perfect AUC returns to leakage and split review; poor missing-input stress returns to feature design; unstable folds return to data and validation design.

Evidence: S4

10. What gets saved, and how could an SLM learn from it?

Layer	What is stored	Why it matters
Raw trial evidence	Result JSON, OOF predictions, recipe, data/code hashes, fold metrics, stress tests, calibration.	Lets an expert reproduce and audit the actual measurement.
Human run summary	Run card, leaderboard, numbered reports, failed attempts.	Lets a person see the answer quickly and drill into details.
Research memory	Cited lessons, counterevidence, limitations, and ten workflow blocks.	Helps future agents retrieve a narrow precedent rather than guessing.
Clean SFT export	Structured chat examples for agenda planning, data/leakage review, experiment proposals, critiques, and synthesis.	A possible training source for a focused small language model after human curation.

The current clean export manifest records 12 completed compact trials and 37 chat-formatted examples across 5 task types. These examples are preparatory. The full trajectory is marked training_ready=false. Before training, verify every claim, remove secrets or personal data, check licenses, eliminate near duplicates, and split by source/dataset or task lineage so related experiments do not leak into evaluation.

Fine tuning could teach a small model to follow a particular review format and recognize common research moves. It cannot guarantee factual memory, remove hallucination by itself, replace fresh data access, or make the whole system deterministic. A useful evaluation should compare the proposed SLM with the present agent plus tools on held-out datasets, citation accuracy, valid plans, error recovery, latency, and total cost.

Where to click in the local Studio

View	Use it for
Research	Set a goal and budget; watch run status, trial scores, critique, and input/output diagnostics.
Knowledge	Read provisional lessons and open the full model building field guide.
Recipes & traces	Inspect replay configuration and export the explicit run trajectory.

Evidence: S8, S9, S10, S13

11. What is proven, and what remains open?

Established in this repository	Still requires new evidence
The studio can coordinate typed agent roles and calculate bounded tabular experiments with preserved local evidence.	That a specialized SLM improves quality, cost, latency, or reliability over the current agent system.
Feature exclusion and fold-local preprocessing are implemented in the worker.	That all source columns are truly available at the real prediction moment.
The controlled 50-task campaign completed and consumed ten final holdouts once.	That its selections hold on new time periods, locations, source systems, and production populations.
The clean export and field guide exist.	Human-reviewed training labels, grouped evaluation sets, and a validated fine-tuned model.
Historical model and feature results are collected.	Reproduction of historical champions with complete data/code/environment provenance.

A sensible next research cycle

- Choose one concrete DCLab workflow that currently takes too much human effort.
- Write a measurable baseline: task completion, citation accuracy, invalid plans, human correction time, latency, and cost.
- Use a new dataset or later time slice for evaluation. Do not train on or repeatedly tune against it.
- Compare current agent plus deterministic worker, a narrower prompt, retrieval, and a small fine-tuned model.
- Promote a capability only when the improvement survives independent testing and the failure modes are understood.

Evidence: S4, S12, S13

Appendix A. Every controlled campaign task

Each dataset has five experiment IDs. The sentences below are taken from the recorded claims and numerical evidence. The final AUC in stage five is a once-consumed holdout; stages one through four use training partition evidence and CV. These rows describe what was measured, not what to deploy.

Evidence: S3

Adult

Five connected tests answer five different questions about the same dataset. Read the ID to find the full JSON evidence.

ID	Question stage	Recorded result in plain words
EXP-001	Data understanding	adult has 48842 source rows, 14 cached features, and a training positive rate of 0.239.
EXP-002	Leakage audit	Declared decision-time exclusions for adult: none.
EXP-003	Feature engineering	Use the raw feature recipe for the next model screen on adult; it achieved 0.8955 mean training-CV ROC-AUC with about 14 features.
EXP-004	Model screen	lightgbm is the training-CV model candidate for adult using raw features (0.8955±0.0065 ROC-AUC across folds).
EXP-005	Optimize and evaluate	The preselected lightgbm/lightgbm_C02 recipe scored 0.9097 ROC-AUC on the once-consumed holdout (bootstrap 95% interval 0.8855–0.9297). Holdout ROC-AUC: 0.9097.

Reading: selected feature recipe = raw; final selected model = lightgbm; holdout Brier = 0.0971. The result is specific to this split and candidate set.

Bank Marketing

Five connected tests answer five different questions about the same dataset. Read the ID to find the full JSON evidence.

ID	Question stage	Recorded result in plain words
EXP-006	Data understanding	bank_marketing has 45211 source rows, 16 cached features, and a training positive rate of 0.117.
EXP-007	Leakage audit	Declared decision-time exclusions for bank_marketing: ['duration']; their apparent training-CV AUC lift was +0.1842.
EXP-008	Feature engineering	Use the ratios feature recipe for the next model screen on bank_marketing; it achieved 0.7172 mean training-CV ROC-AUC with about 38 features.
EXP-009	Model screen	extra_trees is the training-CV model candidate for bank_marketing using ratios features (0.7288±0.0295 ROC-AUC across folds).
EXP-010	Optimize and evaluate	The preselected extra_trees/extra_trees_C02 recipe scored 0.7718 ROC-AUC on the once-consumed holdout (bootstrap 95% interval 0.7126–0.8290). Holdout ROC-AUC: 0.7718.

Reading: selected feature recipe = ratios; final selected model = extra_trees; holdout Brier = 0.0870. The result is specific to this split and candidate set.

Breast Cancer

Five connected tests answer five different questions about the same dataset. Read the ID to find the full JSON evidence.

ID	Question stage	Recorded result in plain words
EXP-011	Data understanding	breast_cancer has 569 source rows, 30 cached features, and a training positive rate of 0.374.
EXP-012	Leakage audit	Declared decision-time exclusions for breast_cancer: none.
EXP-013	Feature engineering	Use the raw feature recipe for the next model screen on breast_cancer; it achieved 0.9907 mean training-CV ROC-AUC with about 30 features.
EXP-014	Model screen	logistic_regression is the training-CV model candidate for breast_cancer using raw features (0.9955±0.0047 ROC-AUC across folds).
EXP-015	Optimize and evaluate	The preselected logistic_regression/baseline recipe scored 0.9960 ROC-AUC on the once-consumed holdout (bootstrap 95% interval 0.9858–1.0000). Holdout ROC-AUC: 0.9960.

Reading: selected feature recipe = raw; final selected model = logistic_regression; holdout Brier = 0.0213. The result is specific to this split and candidate set.

Heart Disease

Five connected tests answer five different questions about the same dataset. Read the ID to find the full JSON evidence.

ID	Question stage	Recorded result in plain words
EXP-016	Data understanding	heart_disease has 303 source rows, 13 cached features, and a training positive rate of 0.459.
EXP-017	Leakage audit	Declared decision-time exclusions for heart_disease: none.
EXP-018	Feature engineering	Use the raw feature recipe for the next model screen on heart_disease; it achieved 0.8700 mean training-CV ROC-AUC with about 13 features.
EXP-019	Model screen	extra_trees is the training-CV model candidate for heart_disease using raw features (0.8978±0.0244 ROC-AUC across folds).
EXP-020	Optimize and evaluate	The preselected extra_trees/extra_trees_C02 recipe scored 0.9448 ROC-AUC on the once-consumed holdout (bootstrap 95% interval 0.8851–0.9872). Holdout ROC-AUC: 0.9448.

Reading: selected feature recipe = raw; final selected model = extra_trees; holdout Brier = 0.1070. The result is specific to this split and candidate set.

Credit Default

Five connected tests answer five different questions about the same dataset. Read the ID to find the full JSON evidence.

ID	Question stage	Recorded result in plain words
EXP-021	Data understanding	credit_default has 30000 source rows, 23 cached features, and a training positive rate of 0.221.
EXP-022	Leakage audit	Declared decision-time exclusions for credit_default: none.
EXP-023	Feature engineering	Use the raw feature recipe for the next model screen on credit_default; it achieved 0.7317 mean training-CV ROC-AUC with about 23 features.
EXP-024	Model screen	extra_trees is the training-CV model candidate for credit_default using raw features (0.7363±0.0159 ROC-AUC across folds).
EXP-025	Optimize and evaluate	The preselected extra_trees/extra_trees_C02 recipe scored 0.7746 ROC-AUC on the once-consumed holdout (bootstrap 95% interval 0.7326–0.8150). Holdout ROC-AUC: 0.7746.

Reading: selected feature recipe = raw; final selected model = extra_trees; holdout Brier = 0.1371. The result is specific to this split and candidate set.

German Credit

Five connected tests answer five different questions about the same dataset. Read the ID to find the full JSON evidence.

ID	Question stage	Recorded result in plain words
EXP-026	Data understanding	german_credit has 1000 source rows, 20 cached features, and a training positive rate of 0.300.
EXP-027	Leakage audit	Declared decision-time exclusions for german_credit: none.
EXP-028	Feature engineering	Use the interactions feature recipe for the next model screen on german_credit; it achieved 0.7601 mean training-CV ROC-AUC with about 56 features.
EXP-029	Model screen	extra_trees is the training-CV model candidate for german_credit using interactions features (0.7769±0.0298 ROC-AUC across folds).
EXP-030	Optimize and evaluate	The preselected extra_trees/extra_trees_C01 recipe scored 0.7943 ROC-AUC on the once-consumed holdout (bootstrap 95% interval 0.7180–0.8621). Holdout ROC-AUC: 0.7943.

Reading: selected feature recipe = interactions; final selected model = extra_trees; holdout Brier = 0.1605. The result is specific to this split and candidate set.

Mushroom

Five connected tests answer five different questions about the same dataset. Read the ID to find the full JSON evidence.

ID	Question stage	Recorded result in plain words
EXP-031	Data understanding	mushroom has 8124 source rows, 22 cached features, and a training positive rate of 0.482.
EXP-032	Leakage audit	Declared decision-time exclusions for mushroom: none.
EXP-033	Feature engineering	Use the raw feature recipe for the next model screen on mushroom; it achieved 1.0000 mean training-CV ROC-AUC with about 22 features.
EXP-034	Model screen	extra_trees is the training-CV model candidate for mushroom using raw features (1.0000±0.0000 ROC-AUC across folds).
EXP-035	Optimize and evaluate	The preselected extra_trees/baseline recipe scored 1.0000 ROC-AUC on the once-consumed holdout (bootstrap 95% interval 1.0000–1.0000). Holdout ROC-AUC: 1.0000.

Reading: selected feature recipe = raw; final selected model = extra_trees; holdout Brier = 0.0000. The result is specific to this split and candidate set.

Spambase

Five connected tests answer five different questions about the same dataset. Read the ID to find the full JSON evidence.

ID	Question stage	Recorded result in plain words
EXP-036	Data understanding	spambase has 4601 source rows, 57 cached features, and a training positive rate of 0.394.
EXP-037	Leakage audit	Declared decision-time exclusions for spambase: none.
EXP-038	Feature engineering	Use the raw feature recipe for the next model screen on spambase; it achieved 0.9869 mean training-CV ROC-AUC with about 57 features.
EXP-039	Model screen	hist_gradient_boosting is the training-CV model candidate for spambase using raw features (0.9874±0.0021 ROC-AUC across folds).
EXP-040	Optimize and evaluate	The preselected hist_gradient_boosting/baseline recipe scored 0.9849 ROC-AUC on the once-consumed holdout (bootstrap 95% interval 0.9752–0.9925). Holdout ROC-AUC: 0.9849.

Reading: selected feature recipe = raw; final selected model = hist_gradient_boosting; holdout Brier = 0.0382. The result is specific to this split and candidate set.

Online Shoppers

Five connected tests answer five different questions about the same dataset. Read the ID to find the full JSON evidence.

ID	Question stage	Recorded result in plain words
EXP-041	Data understanding	online_shoppers has 12330 source rows, 17 cached features, and a training positive rate of 0.155.
EXP-042	Leakage audit	Declared decision-time exclusions for online_shoppers: ['PageValues']; their apparent training-CV AUC lift was +0.1968.
EXP-043	Feature engineering	Use the raw feature recipe for the next model screen on online_shoppers; it achieved 0.7156 mean training-CV ROC-AUC with about 16 features.
EXP-044	Model screen	extra_trees is the training-CV model candidate for online_shoppers using raw features (0.7366±0.0072 ROC-AUC across folds).
EXP-045	Optimize and evaluate	The preselected extra_trees/extra_trees_C02 recipe scored 0.7394 ROC-AUC on the once-consumed holdout (bootstrap 95% interval 0.6997–0.7828). Holdout ROC-AUC: 0.7394.

Reading: selected feature recipe = raw; final selected model = extra_trees; holdout Brier = 0.1182. The result is specific to this split and candidate set.

Wine Quality

Five connected tests answer five different questions about the same dataset. Read the ID to find the full JSON evidence.

ID	Question stage	Recorded result in plain words
EXP-046	Data understanding	wine_quality has 6497 source rows, 11 cached features, and a training positive rate of 0.633.
EXP-047	Leakage audit	Declared decision-time exclusions for wine_quality: none.
EXP-048	Feature engineering	Use the ratios feature recipe for the next model screen on wine_quality; it achieved 0.8429 mean training-CV ROC-AUC with about 32 features.
EXP-049	Model screen	extra_trees is the training-CV model candidate for wine_quality using ratios features (0.8609±0.0036 ROC-AUC across folds).
EXP-050	Optimize and evaluate	The preselected extra_trees/baseline recipe scored 0.8878 ROC-AUC on the once-consumed holdout (bootstrap 95% interval 0.8610–0.9077). Holdout ROC-AUC: 0.8878.

Reading: selected feature recipe = ratios; final selected model = extra_trees; holdout Brier = 0.1319. The result is specific to this split and candidate set.

Appendix B. All 15 fixed churn experiments

These 15 recipes share one development CV study. The AUC change is relative to CHURN-002, the plain logistic baseline. Lower log loss is better; the highest AUC does not decide the best business threshold.

ID	Recipe	Family	AUC	AP	Log loss	Delta AUC
CHURN-014	Logistic with charge features	logistic_regression	0.8499	0.6708	0.4108	0.0050
CHURN-010	Regularized histogram boosting	hist_gradient_boosting	0.8478	0.6670	0.4133	0.0030
CHURN-008	Regularized Random Forest	random_forest	0.8472	0.6623	0.4153	0.0024
CHURN-012	Regularized LightGBM	lightgbm	0.8468	0.6666	0.4143	0.0020
CHURN-015	LightGBM with charge features	lightgbm	0.8466	0.6633	0.4147	0.0018
CHURN-013	XGBoost baseline	xgboost	0.8460	0.6666	0.4152	0.0011
CHURN-004	Flexible logistic C=3	logistic_regression	0.8449	0.6534	0.4181	0.0001
CHURN-002	Logistic baseline C=1	logistic_regression	0.8448	0.6535	0.4181	0.0000
CHURN-003	Regularized logistic C=0.2	logistic_regression	0.8445	0.6529	0.4182	-0.0004
CHURN-011	LightGBM baseline	lightgbm	0.8402	0.6578	0.4236	-0.0046
CHURN-006	Regularized Extra Trees	extra_trees	0.8397	0.6418	0.4239	-0.0052
CHURN-009	Histogram gradient boosting	hist_gradient_boosting	0.8382	0.6540	0.4274	-0.0066
CHURN-007	Random Forest baseline	random_forest	0.8364	0.6443	0.4360	-0.0085
CHURN-005	Extra Trees baseline	extra_trees	0.8246	0.6134	0.4421	-0.0202
CHURN-001	Prior-only floor	dummy	0.5000	0.2654	0.5787	-0.3448

Evidence: S5

Appendix C. Research Studio runs and individual trials

This is the current local ledger, which is more authoritative than an older generated report snapshot. Development CV uses repeated duplicate-group folds. A run can contain successful measured trials even when its overall status is budget_exhausted. Model labels are shortened here; the full recipe is stored under each trial ID.

Run	Dataset	Status	Trials	LLM calls	Tokens
6141d054	telco_churn	budget_exhausted	3	11	81970
cde300a2	adult	failed	0	1	unreported
72a082af	german_credit	completed	4	11	153096
eb59895d	bank_marketing	completed	3	9	125696
d311bd62	bank_marketing	completed	3	9	77318

Run 6141d054 - telco_churn

Trial	Status	Question / change	Model	AUC	AP	Loss	Delta
trial-001	completed	Transparent logistic-regression baseline with billing-feature stress	logistic	0.8447	0.6624	0.4199	-
trial-002	completed	Single-feature ablation: log-transformed tenure in logistic regression	logistic	0.8483	0.6782	0.4142	0.0036
trial-003	completed	Histogram gradient boosting versus log-tenure logistic regression	hist_boost	0.8214	0.6278	0.4694	-0.0270

Run cde300a2 - adult

No model trial was completed. The available data profile and failure event remain in the trajectory.

Run 72a082af - german_credit

Trial	Status	Question / change	Model	AUC	AP	Loss	Delta
trial-001	completed	Original-feature logistic regression baseline	logistic	0.7708	0.5951	0.5233	-
trial-002	completed	Isolated log credit-amount transformation versus logistic baseline	logistic	0.7671	0.5913	0.5262	-0.0037
trial-003	completed	Original-feature random forest versus logistic regression	random_forest	0.7875	0.6380	0.4972	0.0167
trial-004	completed	Random-forest leaf-size ablation: minimum leaf size 5 versus 1	random_forest	0.7917	0.6383	0.5060	0.0041

Run eb59895d - bank_marketing

Trial	Status	Question / change	Model	AUC	AP	Loss	Delta
trial-001	completed	Pre-call logistic baseline with balance and history absence diagnostics	logistic	0.6965	0.3240	0.3000	-
trial-002	completed	Fixed random forest versus the pre-call logistic baseline	random_forest	0.6925	0.3522	0.2924	-0.0039
trial-003	completed	Signed-log balance replacement in the unchanged logistic baseline	logistic	0.7059	0.3283	0.2983	0.0134

Run d311bd62 - bank_marketing

Trial	Status	Question / change	Model	AUC	AP	Loss	Delta
trial-001	completed	Pre-call logistic baseline with balance and history absence stresses	logistic	0.6965	0.3240	0.3000	-
trial-002	completed	Paired random-forest comparison against the pre-call logistic baseline	random_forest	0.6925	0.3522	0.2924	-
trial-003	failed	Final ablation: replace raw balance with signed-log balance in logistic regression	logistic	-	-	-	-

For German Credit, trial-004 improved AUC relative to trial-003 but worsened log loss. For Bank Marketing, the forest's AUC dipped while AP and log loss improved. These disagreements are why one metric does not settle the model choice. One Bank Marketing trial failed, and the adult agentic run stopped before any model trial. The churn agentic run kept three measured trials despite the time budget ending.

Evidence: S6, S8

Appendix D. HyperAck historical experiment index

The 127 registry rows below include 83 core lifecycle records (80 completed, 3 skipped) and 44 four way benchmark rows. This is an index to historical measurements; the rows share data and often the same split, so they are not 127 independent confirmations. 'Skipped' means no measured score was produced for that record.

Evidence: S2, S7

HyperAck Initial (15 rows)

Experiment ID	Model	Mode	AUC	Recall	Status
14	lightgbm	safe	0.9387	0.8025	completed
01	xgboost	unsafe	0.9730	0.8931	completed
02	lightgbm	unsafe	0.9720	0.8805	completed
03	lightgbm	unsafe	0.9767	0.8998	completed
04	lightgbm	unsafe	0.9740	0.8979	completed
05	lightgbm	unsafe	0.9682	0.8796	completed
06	lightgbm	unsafe	0.9778	0.9085	completed
07	lightgbm	unsafe	0.9689	0.8786	completed
08	xgboost	unsafe	0.9788	0.9104	completed
09	lightgbm	unsafe	0.9793	0.9114	completed
10	hist_gradient_boosting	unsafe	0.9776	0.9123	completed
11	tabpfn	unsafe	-	-	skipped
12	lightgbm	unsafe	0.9780	0.8815	completed
13	ensemble	unsafe	0.9737	0.9143	completed
15	ensemble	unsafe	0.9781	0.9114	completed

HyperAck Safe (15 rows)

Experiment ID	Model	Mode	AUC	Recall	Status
01	xgboost	safe	0.9386	0.8025	completed
02	lightgbm	safe	0.9365	0.8035	completed
03	lightgbm	safe	0.9358	0.7948	completed
04	lightgbm	safe	0.9419	0.8150	completed
05	lightgbm	safe	0.9353	0.7919	completed
06	lightgbm	safe	0.9389	0.7987	completed
07	lightgbm	safe	0.9387	0.8064	completed
08	xgboost	safe	0.9419	0.7755	completed
09	lightgbm	safe	0.9450	0.7707	completed
10	hist_gradient_boosting	safe	0.9390	0.8131	completed
11	tabpfn	safe	-	-	skipped
12	lightgbm	safe	0.9386	0.7659	completed
13	ensemble	safe	0.9367	0.8054	completed
14	lightgbm	safe	0.9387	0.8025	completed
15	ensemble	safe	0.9384	0.8054	completed

HyperAck Optimized Safe (53 rows)

Experiment ID	Model	Mode	AUC	Recall	Status
01	lightgbm	safe	0.9381	0.8112	completed
02	lightgbm	safe	0.9372	0.7977	completed
03	lightgbm	safe	0.9360	0.7987	completed
04	lightgbm	safe	0.9362	0.7987	completed
05	xgboost	safe	0.9360	0.8025	completed
06	lightgbm	safe	0.9371	0.8006	completed
07	lightgbm	safe	0.9333	0.8015	completed
08	xgboost	safe	0.9410	0.7697	completed
09	lightgbm	safe	0.9409	0.7871	completed
10	hist_gradient_boosting	safe	0.9366	0.8083	completed
11	tabpfn	safe	-	-	skipped
12	lightgbm	safe	0.9375	0.9374	completed
13	ensemble	safe	0.9364	0.8141	completed
14	lightgbm	safe	0.9366	0.8015	completed
15	ensemble	safe	0.9368	0.8025	completed
16	lightgbm	safe	0.9430	0.7890	completed
17	xgboost	safe	0.9414	0.7890	completed
18	ensemble	safe	0.9433	0.7909	completed
19	ensemble	safe	0.9426	0.7919	completed
20	lightgbm	safe	0.9425	0.7871	completed
21	lightgbm	safe	0.9450	0.7707	completed
22	lightgbm	safe	0.9433	0.7871	completed
23	ensemble	safe	0.9444	0.7649	completed
24	ensemble	safe	0.9443	0.7803	completed
25	lightgbm	safe	0.9404	0.7746	completed
26	lightgbm	safe	0.9432	0.7726	completed
27	ensemble	safe	0.9445	0.7649	completed
28	ensemble	safe	0.9439	0.7688	completed
29	ensemble	safe	0.9430	0.7572	completed
30	catboost	safe	0.9428	0.8025	completed
31	lightgbm	safe	0.9425	0.7775	completed
32	ensemble	safe	0.9443	0.7977	completed
33	ensemble	safe	0.9426	0.7572	completed
34	lightgbm	safe	0.9446	0.7678	completed
35	ensemble	safe	0.9446	0.7688	completed

Experiment ID	Model	Mode	AUC	Recall	Status
36	ensemble	safe	0.9446	0.7649	completed
37	ensemble	safe	0.9438	0.7640	completed
38	ensemble	safe	0.9438	0.7871	completed
39	ensemble	safe	0.9441	0.7688	completed
40	ensemble	safe	0.9443	0.7697	completed
41	logistic_regression	safe	0.8823	0.7187	completed
42	random_forest	safe	0.9414	0.7746	completed
43	extra_trees	safe	0.9444	0.7794	completed
44	hist_gradient_boosting	safe	0.9412	0.7958	completed
45	linear_svc	safe	0.8829	0.7187	completed
46	lightgbm	safe	0.9427	0.7852	completed
47	xgboost	safe	0.9417	0.7900	completed
48	catboost	safe	0.9417	0.7842	completed
49	lightgbm	safe	0.9425	0.7929	completed
50	ensemble	safe	0.9445	0.7871	completed
51	ensemble	safe	0.9450	0.8006	completed
52	ensemble	safe	0.9442	0.7784	completed
53	ensemble	safe	0.9455	0.7784	completed

HyperAck 4Way (44 rows)

Experiment ID	Model	Mode	AUC	Recall	Status
safe_baseline_catboost	catboost	safe	0.9435	0.7938	completed
safe_baseline_extra_trees	extra_trees	safe	0.9368	0.7987	completed
safe_baseline_hist_gradient_boosting	hist_gradient_boosting	safe	0.9426	0.7890	completed
safe_baseline_lightgbm	lightgbm	safe	0.9426	0.7861	completed
safe_baseline_linear_svc	linear_svc	safe	0.8824	0.7187	completed
safe_baseline_logistic_regression	logistic_regression	safe	0.8818	0.7197	completed
safe_baseline_random_forest	random_forest	safe	0.9408	0.7967	completed
safe_baseline_soft_vote_blend	ensemble	safe	0.9442	0.7688	completed
safe_baseline_stacking_ensemble	ensemble	safe	0.9445	0.7929	completed
safe_baseline_tabular_transformer	tabular_transformer	safe	0.9051	0.7601	completed
safe_baseline_xgboost	xgboost	safe	0.9347	0.8160	completed
safe_optimized_catboost	catboost	safe	0.9421	0.7938	completed
safe_optimized_extra_trees	extra_trees	safe	0.9397	0.7563	completed
safe_optimized_hist_gradient_boosting	hist_gradient_boosting	safe	0.9376	0.8035	completed
safe_optimized_lightgbm	lightgbm	safe	0.9449	0.7697	completed
safe_optimized_linear_svc	linear_svc	safe	0.8824	0.7177	completed
safe_optimized_logistic_regression	logistic_regression	safe	0.8818	0.7197	completed
safe_optimized_random_forest	random_forest	safe	0.9413	0.7803	completed
safe_optimized_soft_vote_blend	ensemble	safe	0.9442	0.7688	completed
safe_optimized_stacking_ensemble	ensemble	safe	0.9445	0.7929	completed
safe_optimized_tabular_transformer	tabular_transformer	safe	0.9186	0.7784	completed
safe_optimized_xgboost	xgboost	safe	0.9403	0.7967	completed
unsafe_baseline_catboost	catboost	unsafe	0.9784	0.9191	completed
unsafe_baseline_extra_trees	extra_trees	unsafe	0.9731	0.9114	completed
unsafe_baseline_hist_gradient_boosting	hist_gradient_boosting	unsafe	0.9802	0.9075	completed
unsafe_baseline_lightgbm	lightgbm	unsafe	0.9793	0.9085	completed
unsafe_baseline_linear_svc	linear_svc	unsafe	0.9274	0.8420	completed
unsafe_baseline_logistic_regression	logistic_regression	unsafe	0.9279	0.8343	completed
unsafe_baseline_random_forest	random_forest	unsafe	0.9751	0.9104	completed
unsafe_baseline_soft_vote_blend	ensemble	unsafe	0.9775	0.9123	completed
unsafe_baseline_stacking_ensemble	ensemble	unsafe	0.9781	0.9191	completed
unsafe_baseline_tabular_transformer	tabular_transformer	unsafe	0.9594	0.8786	completed
unsafe_baseline_xgboost	xgboost	unsafe	0.9753	0.9085	completed
unsafe_optimized_catboost	catboost	unsafe	0.9784	0.9171	completed
unsafe_optimized_extra_trees	extra_trees	unsafe	0.9695	0.9008	completed

Experiment ID	Model	Mode	AUC	Recall	Status
unsafe_optimized_hist_gradient_boosting	hist_gradient_boosting	unsafe	0.9773	0.9094	completed
unsafe_optimized_lightgbm	lightgbm	unsafe	0.9789	0.9094	completed
unsafe_optimized_linear_svc	linear_svc	unsafe	0.9275	0.8439	completed
unsafe_optimized_logistic_regression	logistic_regression	unsafe	0.9280	0.8353	completed
unsafe_optimized_random_forest	random_forest	unsafe	0.9753	0.9114	completed
unsafe_optimized_soft_vote_blend	ensemble	unsafe	0.9775	0.9123	completed
unsafe_optimized_stacking_ensemble	ensemble	unsafe	0.9781	0.9191	completed
unsafe_optimized_tabular_transformer	tabular_transformer	unsafe	0.9647	0.9017	completed
unsafe_optimized_xgboost	xgboost	unsafe	0.9786	0.9114	completed

Appendix E. The rest of the historical registry

The normalized registry contains 579 rows in seven suites. The table shows counts and the highest recorded AUC by dataset, including unsafe rows where present; it is not a safe-model leaderboard. Older suites used different experimental setups. The full 579-row lookup, including source path, model, status, and metrics, is knowledge/experiment_registry.csv.

Dataset	Rows	Best registry AUC	Best family
adult	45	0.9274	catboost
bank_marketing	45	0.9386	catboost
breast_cancer	46	1.0000	lightgbm
credit_default	45	0.8005	ensemble
german_credit	45	0.8112	catboost
heart_disease	46	0.9740	hist_gradient_boosting
hyperack	127	0.9802	hist_gradient_boosting
mushroom	45	1.0000	xgboost
online_shoppers	45	0.9324	lightgbm
spambase	45	0.9878	lightgbm
wine_quality	45	0.9160	extra_trees

The 50-task campaign, churn suite, and agentic trials receive detailed individual tables above. Appendix F lists the remaining 452 historical registry rows, giving this PDF an index of all 579 normalized records. Older results should be treated as candidate hypotheses until replicated with full provenance and a fresh evaluation design.

Evidence: S2, S11

Appendix F. Remaining historical experiment index

The following 452 records complete the 579-row normalized registry index alongside Appendix D's 127 HyperAck rows. This compact table shows a subset of fields. The registry CSV holds the full source path, other metrics, provenance flags, and notes for each row. Safe and unsafe records can appear together, so read Mode before interpreting AUC.

Evidence: S2

Adult (45 registry rows)

Suite	Experiment ID	Model	Mode	AUC	F1	Status
external_benchmark	adult__baseline__catboost	catboost	external	0.9268	0.7037	completed
external_benchmark	adult__baseline__extra_trees	extra_trees	external	0.8956	0.6611	completed
external_benchmark	adult__baseline__hist_gradient_boosting	hist_boost	external	0.9250	0.7026	completed
external_benchmark	adult__baseline__lightgbm	lightgbm	external	0.9261	0.7073	completed
external_benchmark	adult__baseline__logistic_regression	logistic	external	0.8502	0.5460	completed
external_benchmark	adult__baseline__random_forest	random_forest	external	0.9067	0.6720	completed
external_benchmark	adult__baseline__xgboost	xgboost	external	0.9224	0.6992	completed
external_benchmark	adult__optimized__catboost	catboost	external	0.9274	0.7046	completed
external_benchmark	adult__optimized__extra_trees	extra_trees	external	0.9074	0.6209	completed
external_benchmark	adult__optimized__hist_gradient_boosting	hist_boost	external	0.9246	0.6991	completed
external_benchmark	adult__optimized__lightgbm	lightgbm	external	0.9237	0.6953	completed
external_benchmark	adult__optimized__logistic_regression	logistic	external	0.8502	0.5460	completed
external_benchmark	adult__optimized__random_forest	random_forest	external	0.9152	0.6886	completed
external_benchmark	adult__optimized__xgboost	xgboost	external	0.9262	0.7067	completed
external_playbook	1	logistic	safe	0.8502	0.5460	completed
external_playbook	1	logistic	unsafe	0.8502	0.5460	completed
external_playbook	10	hist_boost	safe	0.9217	0.7009	completed
external_playbook	10	hist_boost	unsafe	0.9217	0.7009	completed
external_playbook	11	extra_trees	safe	0.9012	0.6329	completed
external_playbook	11	extra_trees	unsafe	0.9012	0.6329	completed
external_playbook	12	lightgbm	safe	0.9235	0.6998	completed
external_playbook	12	lightgbm	unsafe	0.9235	0.6998	completed
external_playbook	13	ensemble	safe	0.9223	0.7040	completed
external_playbook	13	ensemble	unsafe	0.9223	0.7040	completed
external_playbook	14	ensemble	safe	0.9197	0.6858	completed
external_playbook	14	ensemble	unsafe	0.9197	0.6858	completed
external_playbook	15	lightgbm	safe	0.9240	0.7041	completed
external_playbook	15	lightgbm	unsafe	0.9240	0.7041	completed
external_playbook	2	lightgbm	safe	0.9261	0.7073	completed
external_playbook	2	lightgbm	unsafe	0.9261	0.7073	completed
external_playbook	3	lightgbm	safe	0.9261	0.7073	completed
external_playbook	3	lightgbm	unsafe	0.9261	0.7073	completed
external_playbook	4	lightgbm	safe	0.9238	0.7001	completed
external_playbook	4	lightgbm	unsafe	0.9238	0.7001	completed
external_playbook	5	lightgbm	safe	0.9230	0.7004	completed

Suite	Experiment ID	Model	Mode	AUC	F1	Status
external_playbook	5	lightgbm	unsafe	0.9230	0.7004	completed
external_playbook	6	lightgbm	safe	0.9228	0.7040	completed
external_playbook	6	lightgbm	unsafe	0.9228	0.7040	completed
external_playbook	7	lightgbm	safe	0.9101	0.6709	completed
external_playbook	7	lightgbm	unsafe	0.9101	0.6709	completed
external_playbook	8	xgboost	safe	0.9255	0.7085	completed
external_playbook	8	xgboost	unsafe	0.9255	0.7085	completed
external_playbook	9	lightgbm	safe	0.9251	0.7058	completed
external_playbook	9	lightgbm	unsafe	0.9251	0.7058	completed
model_building_50_v1	EXP-005_adult_optimization_reliability	lightgbm	safe	0.9097	0.6833	completed

Bank Marketing (45 registry rows)

Suite	Experiment ID	Model	Mode	AUC	F1	Status
external_benchmark	bank_marketing__baseline__catboost	catboost	external	0.9381	0.5309	completed
external_benchmark	bank_marketing__baseline__extra_trees	extra_trees	external	0.9246	0.4570	completed
external_benchmark	bank_marketing__baseline__hist_gradient_boosting	hist_boost	external	0.9319	0.5457	completed
external_benchmark	bank_marketing__baseline__lightgbm	lightgbm	external	0.9318	0.5562	completed
external_benchmark	bank_marketing__baseline__logistic_regression	logistic	external	0.8910	0.4118	completed
external_benchmark	bank_marketing__baseline__random_forest	random_forest	external	0.9299	0.5189	completed
external_benchmark	bank_marketing__baseline__xgboost	xgboost	external	0.9281	0.5476	completed
external_benchmark	bank_marketing__optimized__catboost	catboost	external	0.9386	0.5345	completed
external_benchmark	bank_marketing__optimized__extra_trees	extra_trees	external	0.9250	0.3012	completed
external_benchmark	bank_marketing__optimized__hist_gradient_boosting	hist_boost	external	0.9336	0.5302	completed
external_benchmark	bank_marketing__optimized__lightgbm	lightgbm	external	0.9328	0.5201	completed
external_benchmark	bank_marketing__optimized__logistic_regression	logistic	external	0.8910	0.4118	completed
external_benchmark	bank_marketing__optimized__random_forest	random_forest	external	0.9327	0.5081	completed
external_benchmark	bank_marketing__optimized__xgboost	xgboost	external	0.9350	0.5432	completed
external_playbook	1	logistic	safe	0.7260	0.2234	completed
external_playbook	1	logistic	unsafe	0.8910	0.4118	completed
external_playbook	10	hist_boost	safe	0.7990	0.3180	completed
external_playbook	10	hist_boost	unsafe	0.9332	0.5546	completed
external_playbook	11	extra_trees	safe	0.7924	0.2876	completed
external_playbook	11	extra_trees	unsafe	0.9283	0.4755	completed
external_playbook	12	lightgbm	safe	0.7930	0.3054	completed
external_playbook	12	lightgbm	unsafe	0.9339	0.5459	completed
external_playbook	13	ensemble	safe	0.7998	0.3380	completed
external_playbook	13	ensemble	unsafe	0.9346	0.5569	completed
external_playbook	14	ensemble	safe	0.8002	0.3139	completed
external_playbook	14	ensemble	unsafe	0.9346	0.5293	completed
external_playbook	15	lightgbm	safe	0.7919	0.3312	completed
external_playbook	15	lightgbm	unsafe	0.9349	0.5479	completed
external_playbook	2	lightgbm	safe	0.8029	0.3333	completed
external_playbook	2	lightgbm	unsafe	0.9318	0.5562	completed
external_playbook	3	lightgbm	safe	0.8029	0.3333	completed
external_playbook	3	lightgbm	unsafe	0.9318	0.5562	completed
external_playbook	4	lightgbm	safe	0.7872	0.3536	completed

Suite	Experiment ID	Model	Mode	AUC	F1	Status
external_playbook	4	lightgbm	unsafe	0.9337	0.5603	completed
external_playbook	5	lightgbm	safe	0.7892	0.3558	completed
external_playbook	5	lightgbm	unsafe	0.9353	0.5687	completed
external_playbook	6	lightgbm	safe	0.7877	0.3502	completed
external_playbook	6	lightgbm	unsafe	0.9337	0.5614	completed
external_playbook	7	lightgbm	safe	0.7837	0.3216	completed
external_playbook	7	lightgbm	unsafe	0.9177	0.5117	completed
external_playbook	8	xgboost	safe	0.7973	0.3105	completed
external_playbook	8	xgboost	unsafe	0.9357	0.5416	completed
external_playbook	9	lightgbm	safe	0.7976	0.3248	completed
external_playbook	9	lightgbm	unsafe	0.9354	0.5486	completed
model_building_50_v1	EXP-010_bank_marketing_optimization_reliability	extra_trees	safe	0.7718	0.2222	completed

Breast Cancer (46 registry rows)

Suite	Experiment ID	Model	Mode	AUC	F1	Status
external_benchmark	breast_cancer__baseline__catboost	catboost	external	0.9987	0.9500	completed
external_benchmark	breast_cancer__baseline__extra_trees	extra_trees	external	0.9987	0.9756	completed
external_benchmark	breast_cancer__baseline__hist_gradient_boosting	hist_boost	external	0.9940	0.9500	completed
external_benchmark	breast_cancer__baseline__lightgbm	lightgbm	external	0.9967	0.9500	completed
external_benchmark	breast_cancer__baseline__logistic_regression	logistic	external	0.9960	0.9512	completed
external_benchmark	breast_cancer__baseline__random_forest	random_forest	external	0.9944	0.9630	completed
external_benchmark	breast_cancer__baseline__xgboost	xgboost	external	0.9940	0.9630	completed
external_benchmark	breast_cancer__optimized__catboost	catboost	external	0.9983	0.9500	completed
external_benchmark	breast_cancer__optimized__extra_trees	extra_trees	external	0.9990	0.9500	completed
external_benchmark	breast_cancer__optimized__hist_gradient_boosting	hist_boost	external	0.9914	0.9630	completed
external_benchmark	breast_cancer__optimized__lightgbm	lightgbm	external	0.9983	0.9630	completed
external_benchmark	breast_cancer__optimized__logistic_regression	logistic	external	0.9960	0.9639	completed
external_benchmark	breast_cancer__optimized__random_forest	random_forest	external	0.9950	0.9630	completed
external_benchmark	breast_cancer__optimized__xgboost	xgboost	external	0.9911	0.9630	completed
external_playbook	1	logistic	safe	0.9960	0.9512	completed
external_playbook	1	logistic	unsafe	0.9960	0.9512	completed
external_playbook	10	hist_boost	safe	0.9970	0.9512	completed
external_playbook	10	hist_boost	unsafe	0.9970	0.9512	completed
external_playbook	11	extra_trees	safe	0.9997	0.9756	completed
external_playbook	11	extra_trees	unsafe	0.9997	0.9756	completed
external_playbook	12	lightgbm	safe	0.9983	0.9512	completed
external_playbook	12	lightgbm	unsafe	0.9983	0.9512	completed
external_playbook	13	ensemble	safe	0.9993	0.9756	completed
external_playbook	13	ensemble	unsafe	0.9993	0.9756	completed
external_playbook	14	ensemble	safe	0.9993	0.9756	completed
external_playbook	14	ensemble	unsafe	0.9993	0.9756	completed
external_playbook	15	lightgbm	safe	0.9980	0.9512	completed
external_playbook	15	lightgbm	unsafe	0.9980	0.9512	completed
external_playbook	2	lightgbm	safe	0.9967	0.9500	completed
external_playbook	2	lightgbm	unsafe	0.9967	0.9500	completed
external_playbook	3	lightgbm	safe	0.9967	0.9500	completed
external_playbook	3	lightgbm	unsafe	0.9967	0.9500	completed
external_playbook	4	lightgbm	safe	1.0000	0.9630	completed
external_playbook	4	lightgbm	unsafe	1.0000	0.9630	completed

Suite	Experiment ID	Model	Mode	AUC	F1	Status
external_playbook	5	lightgbm	safe	0.9987	0.9512	completed
external_playbook	5	lightgbm	unsafe	0.9987	0.9512	completed
external_playbook	6	lightgbm	safe	0.9987	0.9512	completed
external_playbook	6	lightgbm	safe	0.9987	0.9512	completed
external_playbook	6	lightgbm	unsafe	0.9987	0.9512	completed
external_playbook	7	lightgbm	safe	0.9937	0.9512	completed
external_playbook	7	lightgbm	unsafe	0.9937	0.9512	completed
external_playbook	8	xgboost	safe	0.9997	0.9630	completed
external_playbook	8	xgboost	unsafe	0.9997	0.9630	completed
external_playbook	9	lightgbm	safe	0.9990	0.9630	completed
external_playbook	9	lightgbm	unsafe	0.9990	0.9630	completed
model_building_50_v1	EXP-015_breast_cancer_optimization_reliability	logistic	safe	0.9960	0.9512	completed

Credit Default (45 registry rows)

Suite	Experiment ID	Model	Mode	AUC	F1	Status
external_benchmark	credit_default__baseline__catboost	catboost	external	0.7897	0.5018	completed
external_benchmark	credit_default__baseline__extra_trees	extra_trees	external	0.7649	0.4717	completed
external_benchmark	credit_default__baseline__hist_gradient_boosting	hist_boost	external	0.7928	0.4912	completed
external_benchmark	credit_default__baseline__lightgbm	lightgbm	external	0.7867	0.4880	completed
external_benchmark	credit_default__baseline__logistic_regression	logistic	external	0.7209	0.3755	completed
external_benchmark	credit_default__baseline__random_forest	random_forest	external	0.7810	0.4932	completed
external_benchmark	credit_default__baseline__xgboost	xgboost	external	0.7712	0.4801	completed
external_benchmark	credit_default__optimized__catboost	catboost	external	0.7907	0.4898	completed
external_benchmark	credit_default__optimized__extra_trees	extra_trees	external	0.7897	0.4197	completed
external_benchmark	credit_default__optimized__hist_gradient_boosting	hist_boost	external	0.7943	0.4907	completed
external_benchmark	credit_default__optimized__lightgbm	lightgbm	external	0.7974	0.4829	completed
external_benchmark	credit_default__optimized__logistic_regression	logistic	external	0.7209	0.3755	completed
external_benchmark	credit_default__optimized__random_forest	random_forest	external	0.7961	0.4790	completed
external_benchmark	credit_default__optimized__xgboost	xgboost	external	0.7879	0.4996	completed
external_playbook	1	logistic	safe	0.7209	0.3755	completed
external_playbook	1	logistic	unsafe	0.7209	0.3755	completed
external_playbook	10	hist_boost	safe	0.7973	0.4874	completed
external_playbook	10	hist_boost	unsafe	0.7973	0.4874	completed
external_playbook	11	extra_trees	safe	0.7957	0.4789	completed
external_playbook	11	extra_trees	unsafe	0.7957	0.4789	completed
external_playbook	12	lightgbm	safe	0.7929	0.4762	completed
external_playbook	12	lightgbm	unsafe	0.7929	0.4762	completed
external_playbook	13	ensemble	safe	0.8005	0.4848	completed
external_playbook	13	ensemble	unsafe	0.8005	0.4848	completed
external_playbook	14	ensemble	safe	0.7996	0.4863	completed
external_playbook	14	ensemble	unsafe	0.7996	0.4863	completed
external_playbook	15	lightgbm	safe	0.7996	0.4866	completed
external_playbook	15	lightgbm	unsafe	0.7996	0.4866	completed
external_playbook	2	lightgbm	safe	0.7867	0.4880	completed
external_playbook	2	lightgbm	unsafe	0.7867	0.4880	completed
external_playbook	3	lightgbm	safe	0.7867	0.4880	completed
external_playbook	3	lightgbm	unsafe	0.7867	0.4880	completed
external_playbook	4	lightgbm	safe	0.7948	0.4900	completed
external_playbook	4	lightgbm	unsafe	0.7948	0.4900	completed

Suite	Experiment ID	Model	Mode	AUC	F1	Status
external_playbook	5	lightgbm	safe	0.7899	0.4870	completed
external_playbook	5	lightgbm	unsafe	0.7899	0.4870	completed
external_playbook	6	lightgbm	safe	0.7930	0.4775	completed
external_playbook	6	lightgbm	unsafe	0.7930	0.4775	completed
external_playbook	7	lightgbm	safe	0.7880	0.4779	completed
external_playbook	7	lightgbm	unsafe	0.7880	0.4779	completed
external_playbook	8	xgboost	safe	0.7988	0.4825	completed
external_playbook	8	xgboost	unsafe	0.7988	0.4825	completed
external_playbook	9	lightgbm	safe	0.7985	0.4802	completed
external_playbook	9	lightgbm	unsafe	0.7985	0.4802	completed
model_building_50_v1	EXP-025_credit_default_optimization_reliability	extra_trees	safe	0.7746	0.4198	completed

German Credit (45 registry rows)

Suite	Experiment ID	Model	Mode	AUC	F1	Status
external_benchmark	german_credit__baseline__catboost	catboost	external	0.8099	0.5714	completed
external_benchmark	german_credit__baseline__extra_trees	extra_trees	external	0.7876	0.5417	completed
external_benchmark	german_credit__baseline__hist_gradient_boosting	hist_boost	external	0.7637	0.5607	completed
external_benchmark	german_credit__baseline__lightgbm	lightgbm	external	0.7827	0.5714	completed
external_benchmark	german_credit__baseline__logistic_regression	logistic	external	0.7748	0.4848	completed
external_benchmark	german_credit__baseline__random_forest	random_forest	external	0.8020	0.5474	completed
external_benchmark	german_credit__baseline__xgboost	xgboost	external	0.7762	0.5818	completed
external_benchmark	german_credit__optimized__catboost	catboost	external	0.8112	0.5962	completed
external_benchmark	german_credit__optimized__extra_trees	extra_trees	external	0.7929	0.3908	completed
external_benchmark	german_credit__optimized__hist_gradient_boosting	hist_boost	external	0.7533	0.5333	completed
external_benchmark	german_credit__optimized__lightgbm	lightgbm	external	0.7798	0.5524	completed
external_benchmark	german_credit__optimized__logistic_regression	logistic	external	0.7746	0.4848	completed
external_benchmark	german_credit__optimized__random_forest	random_forest	external	0.8035	0.5625	completed
external_benchmark	german_credit__optimized__xgboost	xgboost	external	0.7732	0.5524	completed
external_playbook	1	logistic	safe	0.7748	0.4848	completed
external_playbook	1	logistic	unsafe	0.7748	0.4848	completed
external_playbook	10	hist_boost	safe	0.7926	0.5688	completed
external_playbook	10	hist_boost	unsafe	0.7926	0.5688	completed
external_playbook	11	extra_trees	safe	0.7985	0.5000	completed
external_playbook	11	extra_trees	unsafe	0.7985	0.5000	completed
external_playbook	12	lightgbm	safe	0.7840	0.5714	completed
external_playbook	12	lightgbm	unsafe	0.7840	0.5714	completed
external_playbook	13	ensemble	safe	0.8062	0.6038	completed
external_playbook	13	ensemble	unsafe	0.8062	0.6038	completed
external_playbook	14	ensemble	safe	0.8050	0.6214	completed
external_playbook	14	ensemble	unsafe	0.8050	0.6214	completed
external_playbook	15	lightgbm	safe	0.7973	0.6306	completed
external_playbook	15	lightgbm	unsafe	0.7973	0.6306	completed
external_playbook	2	lightgbm	safe	0.7827	0.5714	completed
external_playbook	2	lightgbm	unsafe	0.7827	0.5714	completed
external_playbook	3	lightgbm	safe	0.7827	0.5714	completed
external_playbook	3	lightgbm	unsafe	0.7827	0.5714	completed
external_playbook	4	lightgbm	safe	0.8051	0.6296	completed
external_playbook	4	lightgbm	unsafe	0.8051	0.6296	completed

Suite	Experiment ID	Model	Mode	AUC	F1	Status
external_playbook	5	lightgbm	safe	0.7958	0.6542	completed
external_playbook	5	lightgbm	unsafe	0.7958	0.6542	completed
external_playbook	6	lightgbm	safe	0.7989	0.6372	completed
external_playbook	6	lightgbm	unsafe	0.7989	0.6372	completed
external_playbook	7	lightgbm	safe	0.7662	0.5138	completed
external_playbook	7	lightgbm	unsafe	0.7662	0.5138	completed
external_playbook	8	xgboost	safe	0.8082	0.5882	completed
external_playbook	8	xgboost	unsafe	0.8082	0.5882	completed
external_playbook	9	lightgbm	safe	0.7905	0.5400	completed
external_playbook	9	lightgbm	unsafe	0.7905	0.5400	completed
model_building_50_v1	EXP-030_german_credit_optimization_reliability	extra_trees	safe	0.7943	0.4848	completed

Heart Disease (46 registry rows)

Suite	Experiment ID	Model	Mode	AUC	F1	Status
external_benchmark	heart_disease__baseline__catboost	catboost	external	0.9556	0.8852	completed
external_benchmark	heart_disease__baseline__extra_trees	extra_trees	external	0.9524	0.8387	completed
external_benchmark	heart_disease__baseline__hist_gradient_boosting	hist_boost	external	0.9621	0.9310	completed
external_benchmark	heart_disease__baseline__lightgbm	lightgbm	external	0.9524	0.8814	completed
external_benchmark	heart_disease__baseline__logistic_regression	logistic	external	0.9513	0.8667	completed
external_benchmark	heart_disease__baseline__random_forest	random_forest	external	0.9578	0.9153	completed
external_benchmark	heart_disease__baseline__xgboost	xgboost	external	0.9188	0.8525	completed
external_benchmark	heart_disease__optimized__catboost	catboost	external	0.9318	0.8814	completed
external_benchmark	heart_disease__optimized__extra_trees	extra_trees	external	0.9470	0.8333	completed
external_benchmark	heart_disease__optimized__hist_gradient_boosting	hist_boost	external	0.9416	0.8667	completed
external_benchmark	heart_disease__optimized__lightgbm	lightgbm	external	0.9567	0.8772	completed
external_benchmark	heart_disease__optimized__logistic_regression	logistic	external	0.9513	0.8667	completed
external_benchmark	heart_disease__optimized__random_forest	random_forest	external	0.9621	0.8966	completed
external_benchmark	heart_disease__optimized__xgboost	xgboost	external	0.9448	0.8814	completed
external_playbook	1	logistic	safe	0.9513	0.8667	completed
external_playbook	1	logistic	unsafe	0.9513	0.8667	completed
external_playbook	10	hist_boost	safe	0.9740	0.9153	completed
external_playbook	10	hist_boost	unsafe	0.9740	0.9153	completed
external_playbook	11	extra_trees	safe	0.9340	0.8070	completed
external_playbook	11	extra_trees	unsafe	0.9340	0.8070	completed
external_playbook	12	lightgbm	safe	0.9535	0.8814	completed
external_playbook	12	lightgbm	unsafe	0.9535	0.8814	completed
external_playbook	13	ensemble	safe	0.9470	0.8621	completed
external_playbook	13	ensemble	unsafe	0.9470	0.8621	completed
external_playbook	14	ensemble	safe	0.9491	0.8475	completed
external_playbook	14	ensemble	unsafe	0.9491	0.8475	completed
external_playbook	15	lightgbm	safe	0.9665	0.8852	completed
external_playbook	15	lightgbm	unsafe	0.9665	0.8852	completed
external_playbook	2	lightgbm	safe	0.9524	0.8814	completed
external_playbook	2	lightgbm	unsafe	0.9524	0.8814	completed
external_playbook	3	lightgbm	safe	0.9524	0.8814	completed
external_playbook	3	lightgbm	unsafe	0.9524	0.8814	completed
external_playbook	4	lightgbm	safe	0.9524	0.8814	completed
external_playbook	4	lightgbm	unsafe	0.9524	0.8814	completed

Suite	Experiment ID	Model	Mode	AUC	F1	Status
external_playbook	5	lightgbm	safe	0.9600	0.8852	completed
external_playbook	5	lightgbm	unsafe	0.9600	0.8852	completed
external_playbook	6	lightgbm	safe	0.9632	0.8852	completed
external_playbook	6	lightgbm	safe	0.9632	0.8852	completed
external_playbook	6	lightgbm	unsafe	0.9632	0.8852	completed
external_playbook	7	lightgbm	safe	0.9502	0.8571	completed
external_playbook	7	lightgbm	unsafe	0.9502	0.8571	completed
external_playbook	8	xgboost	safe	0.9535	0.8814	completed
external_playbook	8	xgboost	unsafe	0.9535	0.8814	completed
external_playbook	9	lightgbm	safe	0.9545	0.8621	completed
external_playbook	9	lightgbm	unsafe	0.9545	0.8621	completed
model_building_50_v1	EXP-020_heart_disease_optimization_reliability	extra_trees	safe	0.9448	0.8136	completed

Mushroom (45 registry rows)

Suite	Experiment ID	Model	Mode	AUC	F1	Status
external_benchmark	mushroom__baseline__catboost	catboost	external	1.0000	1.0000	completed
external_benchmark	mushroom__baseline__extra_trees	extra_trees	external	1.0000	1.0000	completed
external_benchmark	mushroom__baseline__hist_gradient_boosting	hist_boost	external	1.0000	1.0000	completed
external_benchmark	mushroom__baseline__lightgbm	lightgbm	external	1.0000	1.0000	completed
external_benchmark	mushroom__baseline__logistic_regression	logistic	external	0.9996	0.9871	completed
external_benchmark	mushroom__baseline__random_forest	random_forest	external	1.0000	1.0000	completed
external_benchmark	mushroom__baseline__xgboost	xgboost	external	1.0000	1.0000	completed
external_benchmark	mushroom__optimized__catboost	catboost	external	1.0000	1.0000	completed
external_benchmark	mushroom__optimized__extra_trees	extra_trees	external	1.0000	1.0000	completed
external_benchmark	mushroom__optimized__hist_gradient_boosting	hist_boost	external	1.0000	1.0000	completed
external_benchmark	mushroom__optimized__lightgbm	lightgbm	external	1.0000	1.0000	completed
external_benchmark	mushroom__optimized__logistic_regression	logistic	external	0.9997	0.9878	completed
external_benchmark	mushroom__optimized__random_forest	random_forest	external	1.0000	1.0000	completed
external_benchmark	mushroom__optimized__xgboost	xgboost	external	1.0000	1.0000	completed
external_playbook	1	logistic	safe	0.9996	0.9871	completed
external_playbook	1	logistic	unsafe	0.9996	0.9871	completed
external_playbook	10	hist_boost	safe	1.0000	1.0000	completed
external_playbook	10	hist_boost	unsafe	1.0000	1.0000	completed
external_playbook	11	extra_trees	safe	1.0000	1.0000	completed
external_playbook	11	extra_trees	unsafe	1.0000	1.0000	completed
external_playbook	12	lightgbm	safe	1.0000	1.0000	completed
external_playbook	12	lightgbm	unsafe	1.0000	1.0000	completed
external_playbook	13	ensemble	safe	1.0000	1.0000	completed
external_playbook	13	ensemble	unsafe	1.0000	1.0000	completed
external_playbook	14	ensemble	safe	1.0000	1.0000	completed
external_playbook	14	ensemble	unsafe	1.0000	1.0000	completed
external_playbook	15	lightgbm	safe	1.0000	1.0000	completed
external_playbook	15	lightgbm	unsafe	1.0000	1.0000	completed
external_playbook	2	lightgbm	safe	1.0000	1.0000	completed
external_playbook	2	lightgbm	unsafe	1.0000	1.0000	completed
external_playbook	3	lightgbm	safe	1.0000	1.0000	completed
external_playbook	3	lightgbm	unsafe	1.0000	1.0000	completed
external_playbook	4	lightgbm	safe	1.0000	1.0000	completed
external_playbook	4	lightgbm	unsafe	1.0000	1.0000	completed

Suite	Experiment ID	Model	Mode	AUC	F1	Status
external_playbook	5	lightgbm	safe	1.0000	1.0000	completed
external_playbook	5	lightgbm	unsafe	1.0000	1.0000	completed
external_playbook	6	lightgbm	safe	1.0000	1.0000	completed
external_playbook	6	lightgbm	unsafe	1.0000	1.0000	completed
external_playbook	7	lightgbm	safe	1.0000	1.0000	completed
external_playbook	7	lightgbm	unsafe	1.0000	1.0000	completed
external_playbook	8	xgboost	safe	1.0000	1.0000	completed
external_playbook	8	xgboost	unsafe	1.0000	1.0000	completed
external_playbook	9	lightgbm	safe	1.0000	1.0000	completed
external_playbook	9	lightgbm	unsafe	1.0000	1.0000	completed
model_building_50_v1	EXP-035_mushroom_optimization_reliability	extra_trees	safe	1.0000	1.0000	completed

Online Shoppers (45 registry rows)

Suite	Experiment ID	Model	Mode	AUC	F1	Status
external_benchmark	online_shoppers__baseline__catboost	catboost	external	0.9281	0.6291	completed
external_benchmark	online_shoppers__baseline__extra_trees	extra_trees	external	0.9109	0.5845	completed
external_benchmark	online_shoppers__baseline__hist_gradient_boosting	hist_boost	external	0.9294	0.6514	completed
external_benchmark	online_shoppers__baseline__lightgbm	lightgbm	external	0.9264	0.6331	completed
external_benchmark	online_shoppers__baseline__logistic_regression	logistic	external	0.8672	0.4516	completed
external_benchmark	online_shoppers__baseline__random_forest	random_forest	external	0.9205	0.6510	completed
external_benchmark	online_shoppers__baseline__xgboost	xgboost	external	0.9201	0.6270	completed
external_benchmark	online_shoppers__optimized__catboost	catboost	external	0.9281	0.6325	completed
external_benchmark	online_shoppers__optimized__extra_trees	extra_trees	external	0.9089	0.4595	completed
external_benchmark	online_shoppers__optimized__hist_gradient_boosting	hist_boost	external	0.9269	0.6371	completed
external_benchmark	online_shoppers__optimized__lightgbm	lightgbm	external	0.9324	0.6667	completed
external_benchmark	online_shoppers__optimized__logistic_regression	logistic	external	0.8672	0.4516	completed
external_benchmark	online_shoppers__optimized__random_forest	random_forest	external	0.9229	0.6379	completed
external_benchmark	online_shoppers__optimized__xgboost	xgboost	external	0.9260	0.6638	completed
external_playbook	1	logistic	safe	0.7048	0.0104	completed
external_playbook	1	logistic	unsafe	0.8672	0.4516	completed
external_playbook	10	hist_boost	safe	0.7696	0.2008	completed
external_playbook	10	hist_boost	unsafe	0.9259	0.6088	completed
external_playbook	11	extra_trees	safe	0.7532	0.0203	completed
external_playbook	11	extra_trees	unsafe	0.9213	0.6136	completed
external_playbook	12	lightgbm	safe	0.7648	0.0103	completed
external_playbook	12	lightgbm	unsafe	0.9266	0.6177	completed
external_playbook	13	ensemble	safe	0.7712	0.2118	completed
external_playbook	13	ensemble	unsafe	0.9286	0.6209	completed
external_playbook	14	ensemble	safe	0.7735	0.0903	completed
external_playbook	14	ensemble	unsafe	0.9293	0.6219	completed
external_playbook	15	lightgbm	safe	0.7714	0.1718	completed
external_playbook	15	lightgbm	unsafe	0.9304	0.6409	completed
external_playbook	2	lightgbm	safe	0.7675	0.1962	completed
external_playbook	2	lightgbm	unsafe	0.9264	0.6331	completed
external_playbook	3	lightgbm	safe	0.7675	0.1962	completed
external_playbook	3	lightgbm	unsafe	0.9264	0.6331	completed
external_playbook	4	lightgbm	safe	0.7612	0.1656	completed

Suite	Experiment ID	Model	Mode	AUC	F1	Status
external_playbook	4	lightgbm	unsafe	0.9278	0.6455	completed
external_playbook	5	lightgbm	safe	0.7687	0.1942	completed
external_playbook	5	lightgbm	unsafe	0.9257	0.6286	completed
external_playbook	6	lightgbm	safe	0.7675	0.1807	completed
external_playbook	6	lightgbm	unsafe	0.9263	0.6293	completed
external_playbook	7	lightgbm	safe	0.7250	0.0507	completed
external_playbook	7	lightgbm	unsafe	0.8846	0.5818	completed
external_playbook	8	xgboost	safe	0.7737	0.1212	completed
external_playbook	8	xgboost	unsafe	0.9299	0.6540	completed
external_playbook	9	lightgbm	safe	0.7719	0.1209	completed
external_playbook	9	lightgbm	unsafe	0.9311	0.6373	completed
model_building_50_v1	EXP-045_online_shoppers_optimization_reliability	extra_trees	safe	0.7394	0.0159	completed

Spambase (45 registry rows)

Suite	Experiment ID	Model	Mode	AUC	F1	Status
external_benchmark	spambase__baseline__catboost	catboost	external	0.9869	0.9378	completed
external_benchmark	spambase__baseline__extra_trees	extra_trees	external	0.9828	0.9397	completed
external_benchmark	spambase__baseline__hist_gradient_boosting	hist_boost	external	0.9869	0.9324	completed
external_benchmark	spambase__baseline__lightgbm	lightgbm	external	0.9878	0.9350	completed
external_benchmark	spambase__baseline__logistic_regression	logistic	external	0.9702	0.9093	completed
external_benchmark	spambase__baseline__random_forest	random_forest	external	0.9834	0.9296	completed
external_benchmark	spambase__baseline__xgboost	xgboost	external	0.9861	0.9352	completed
external_benchmark	spambase__optimized__catboost	catboost	external	0.9874	0.9366	completed
external_benchmark	spambase__optimized__extra_trees	extra_trees	external	0.9723	0.8789	completed
external_benchmark	spambase__optimized__hist_gradient_boosting	hist_boost	external	0.9864	0.9354	completed
external_benchmark	spambase__optimized__lightgbm	lightgbm	external	0.9864	0.9265	completed
external_benchmark	spambase__optimized__logistic_regression	logistic	external	0.9704	0.9093	completed
external_benchmark	spambase__optimized__random_forest	random_forest	external	0.9820	0.9187	completed
external_benchmark	spambase__optimized__xgboost	xgboost	external	0.9869	0.9306	completed
external_playbook	1	logistic	safe	0.9702	0.9093	completed
external_playbook	1	logistic	unsafe	0.9702	0.9093	completed
external_playbook	10	hist_boost	safe	0.9857	0.9326	completed
external_playbook	10	hist_boost	unsafe	0.9857	0.9326	completed
external_playbook	11	extra_trees	safe	0.9779	0.9135	completed
external_playbook	11	extra_trees	unsafe	0.9779	0.9135	completed
external_playbook	12	lightgbm	safe	0.9853	0.9368	completed
external_playbook	12	lightgbm	unsafe	0.9853	0.9368	completed
external_playbook	13	ensemble	safe	0.9869	0.9376	completed
external_playbook	13	ensemble	unsafe	0.9869	0.9376	completed
external_playbook	14	ensemble	safe	0.9859	0.9369	completed
external_playbook	14	ensemble	unsafe	0.9859	0.9369	completed
external_playbook	15	lightgbm	safe	0.9877	0.9350	completed
external_playbook	15	lightgbm	unsafe	0.9877	0.9350	completed
external_playbook	2	lightgbm	safe	0.9878	0.9350	completed
external_playbook	2	lightgbm	unsafe	0.9878	0.9350	completed
external_playbook	3	lightgbm	safe	0.9878	0.9350	completed
external_playbook	3	lightgbm	unsafe	0.9878	0.9350	completed
external_playbook	4	lightgbm	safe	0.9863	0.9284	completed
external_playbook	4	lightgbm	unsafe	0.9863	0.9284	completed

Suite	Experiment ID	Model	Mode	AUC	F1	Status
external_playbook	5	lightgbm	safe	0.9865	0.9350	completed
external_playbook	5	lightgbm	unsafe	0.9865	0.9350	completed
external_playbook	6	lightgbm	safe	0.9857	0.9293	completed
external_playbook	6	lightgbm	unsafe	0.9857	0.9293	completed
external_playbook	7	lightgbm	safe	0.9448	0.8773	completed
external_playbook	7	lightgbm	unsafe	0.9448	0.8773	completed
external_playbook	8	xgboost	safe	0.9866	0.9335	completed
external_playbook	8	xgboost	unsafe	0.9866	0.9335	completed
external_playbook	9	lightgbm	safe	0.9856	0.9381	completed
external_playbook	9	lightgbm	unsafe	0.9856	0.9381	completed
model_building_50_v1	EXP-040_spambase_optimization_reliability	hist_boost	safe	0.9849	0.9392	completed

Wine Quality (45 registry rows)

Suite	Experiment ID	Model	Mode	AUC	F1	Status
external_benchmark	wine_quality__baseline__catboost	catboost	external	0.8730	0.8481	completed
external_benchmark	wine_quality__baseline__extra_trees	extra_trees	external	0.9160	0.8833	completed
external_benchmark	wine_quality__baseline__hist_gradient_boosting	hist_boost	external	0.8786	0.8637	completed
external_benchmark	wine_quality__baseline__lightgbm	lightgbm	external	0.8778	0.8555	completed
external_benchmark	wine_quality__baseline__logistic_regression	logistic	external	0.8045	0.8002	completed
external_benchmark	wine_quality__baseline__random_forest	random_forest	external	0.9068	0.8799	completed
external_benchmark	wine_quality__baseline__xgboost	xgboost	external	0.8843	0.8583	completed
external_benchmark	wine_quality__optimized__catboost	catboost	external	0.8810	0.8560	completed
external_benchmark	wine_quality__optimized__extra_trees	extra_trees	external	0.8667	0.8481	completed
external_benchmark	wine_quality__optimized__hist_gradient_boosting	hist_boost	external	0.8879	0.8688	completed
external_benchmark	wine_quality__optimized__lightgbm	lightgbm	external	0.8560	0.8375	completed
external_benchmark	wine_quality__optimized__logistic_regression	logistic	external	0.8045	0.8002	completed
external_benchmark	wine_quality__optimized__random_forest	random_forest	external	0.8884	0.8581	completed
external_benchmark	wine_quality__optimized__xgboost	xgboost	external	0.8844	0.8604	completed
external_playbook	1	logistic	safe	0.8045	0.8002	completed
external_playbook	1	logistic	unsafe	0.8045	0.8002	completed
external_playbook	10	hist_boost	safe	0.8831	0.8546	completed
external_playbook	10	hist_boost	unsafe	0.8831	0.8546	completed
external_playbook	11	extra_trees	safe	0.8844	0.8580	completed
external_playbook	11	extra_trees	unsafe	0.8844	0.8580	completed
external_playbook	12	lightgbm	safe	0.8929	0.8663	completed
external_playbook	12	lightgbm	unsafe	0.8929	0.8663	completed
external_playbook	13	ensemble	safe	0.8968	0.8729	completed
external_playbook	13	ensemble	unsafe	0.8968	0.8729	completed
external_playbook	14	ensemble	safe	0.8831	0.8576	completed
external_playbook	14	ensemble	unsafe	0.8831	0.8576	completed
external_playbook	15	lightgbm	safe	0.8695	0.8448	completed
external_playbook	15	lightgbm	unsafe	0.8695	0.8448	completed
external_playbook	2	lightgbm	safe	0.8778	0.8555	completed
external_playbook	2	lightgbm	unsafe	0.8778	0.8555	completed
external_playbook	3	lightgbm	safe	0.8778	0.8555	completed
external_playbook	3	lightgbm	unsafe	0.8778	0.8555	completed
external_playbook	4	lightgbm	safe	0.8816	0.8531	completed
external_playbook	4	lightgbm	unsafe	0.8816	0.8531	completed

Suite	Experiment ID	Model	Mode	AUC	F1	Status
external_playbook	5	lightgbm	safe	0.8877	0.8593	completed
external_playbook	5	lightgbm	unsafe	0.8877	0.8593	completed
external_playbook	6	lightgbm	safe	0.8860	0.8590	completed
external_playbook	6	lightgbm	unsafe	0.8860	0.8590	completed
external_playbook	7	lightgbm	safe	0.8680	0.8393	completed
external_playbook	7	lightgbm	unsafe	0.8680	0.8393	completed
external_playbook	8	xgboost	safe	0.8946	0.8711	completed
external_playbook	8	xgboost	unsafe	0.8946	0.8711	completed
external_playbook	9	lightgbm	safe	0.8909	0.8652	completed
external_playbook	9	lightgbm	unsafe	0.8909	0.8652	completed
model_building_50_v1	EXP-050_wine_quality_optimization_reliability	extra_trees	safe	0.8878	0.8577	completed

Sources and evidence map

Repository paths below are relative to the R&D checkout. The DCLab product context lives in a separate local checkout and supplies product intent; run metrics come from this R&D repository.

ID	Source	Used for
S1	/Users/shahriar/Downloads/decision_ai/DCLAB_MASTER_CONTEXT.md	Main DCLab concept and separation from the R&D repository.
S2	knowledge/KNOWLEDGE_BASE.md; knowledge/evidence.json; knowledge/experiment_registry.csv	579-row historical registry and cross-dataset/leakage summaries.
S3	campaigns/model_building_50_v1/CAMPAIGN_REPORT.md; results/EXP-*.json	All 50 controlled tasks and ten holdout outcomes.
S4	knowledge/MODEL_BUILDING_FIELD_GUIDE.md; knowledge/model_building_rules.jsonl; knowledge/workflow_blocks.json	Rules, limitations, and ten workflow blocks.
S5	churn_exp/CHURN_BENCHMARK.md; churn_exp/results/summary.json; CHURN-*/result.json	All 15 fixed churn results and protocol.
S6	agent_runs/research.sqlite3; agent_runs/<run_id>/trajectory.json	Current agentic run status, trials, critiques, and usage.
S7	hyperack_exp/; safe_leakage_exp/; optimized_safe_model/; general_pipeline/results/	HyperAck 83-test lifecycle and 44-row four way benchmark.
S8	dclab_rnd/agentic/agents.py; engine.py; worker.py; schemas.py	Agent roles, graph sequence, bounded recipe, evaluator.
S9	dclab_rnd/agentic/store.py; archive.py; server.py	Evidence store, archive, local API and UI.
S10	dclab_rnd/agentic/static/index.html; app.js	Research, Knowledge, and Recipes interface.
S11	knowledge/DATA_QUALITY.md; knowledge/NEXT_EXPERIMENTS.md	Registry quality caveats and evidence gaps.
S12	DCLAB_intiatl_MASTER_CONTEXT.md	Initial product vision, treated as older context.
S13	agent_runs/clean_exports/MANIFEST.json; README.md	Current compact trial and SFT export inventory.

A practical reading order: start with this PDF, open the Field Guide for methodology, inspect a campaign EXP JSON or Studio trial result for a claim, then check its recipe, split, code/data hashes, and limitations before reuse.